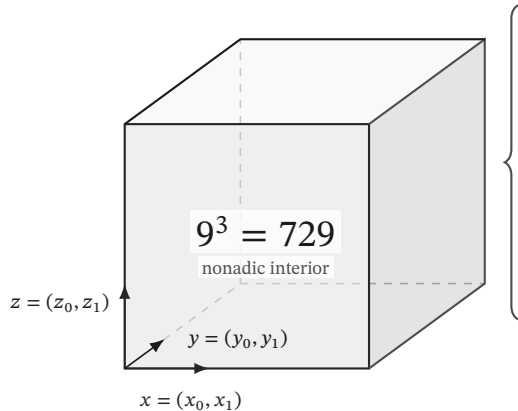




paired ternary coordinates in
three independent directions

side length
9 to 10
shell 271

Cubic completion

interior: 729

new faces
 $243 = 3 \cdot 9^2$

edges
 $27 = 3 \cdot 9$

vertex
1

$$1000 = 729 + 243 + 27 + 1 = 729 + 271$$

inner boundary of the cube of side 9: 386 points

shell of the completion $9^3 \subset 10^3$: 271 states